

Kartik Sisodia

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EDUCATION

Delhi Technological University

Bachelor of Technology in Software Engineering — CGPA: 8.1

Delhi, India

2025 – 2029

EXPERIENCE

Python Developer Intern

Jul 2026 – Oct 2026

Axlero Solutions

Remote

- Developing **VisionEdge**, a hardware-accelerated computer vision pipeline for real-time object detection on high-resolution RTSP video streams using TensorRT, CuPy, and WebRTC.
- Building GPU-accelerated inference pipelines with ONNX, TensorRT, and asynchronous Python to optimize throughput and reduce inference latency.
- Implementing scalable backend components for video streaming, model deployment, and real-time telemetry while collaborating in an Agile development environment.
- Working with modern backend technologies, code reviews, and performance optimization techniques to deliver production-ready Python systems.

PROJECTS

AI-Based Ground-Level Air Quality Estimation | Python, XGBoost, INSAT-3DR, ERA5

- Developed a machine learning framework to estimate six ground-level air pollutants using satellite observations, meteorological variables, and CPCB monitoring data.
- Achieved an overall R^2 score of 0.84 using a Multi-Output XGBoost regression model after evaluating Random Forest and CNN-LSTM approaches.
- Integrated, cleaned, and preprocessed INSAT-3DR, ERA5, and CPCB datasets, creating a modelling dataset of 400+ samples with 15 engineered environmental features.
- Performed feature importance analysis, HCHO hotspot detection, and atmospheric transport analysis to improve model interpretability.

Legal.AI | Python, FastAPI, LangGraph, ChromaDB, Gemma 4, React | GitHub

- Built a multilingual AI-powered legal assistant that delivers citation-backed responses using a hierarchical Retrieval-Augmented Generation (RAG) pipeline over Indian legal statutes.
- Designed and implemented the backend architecture, including document ingestion, intelligent chunking, metadata extraction, semantic indexing, and hierarchical Act → Section retrieval using ChromaDB and Sentence Transformers.
- Developed a modular LangGraph workflow integrating Gemma 4 with streaming Server-Sent Events (SSE), multilingual query processing, and citation-grounded response generation to improve factual reliability.
- Led backend development and AI pipeline design for a 4-member team, analyzing 7+ legal research papers and securing **3rd Place** at Google's Build with Gemma Hackathon (AI for Legal Assistance Track).

College Track | HTML, CSS, JavaScript, Firebase, Cloud Firestore, PWA | GitHub | Live Demo

- Developed a Progressive Web Application for managing attendance, assignments, and academic schedules with real-time cloud synchronization.
- Implemented Firebase Authentication and Cloud Firestore for secure user management and persistent cloud-based data storage.
- Built attendance analytics, assignment tracking, timetable management, guest mode, and responsive interfaces for desktop and mobile devices.
- Architected and developed the complete application, including frontend, database design, authentication, and deployment.

JOLT | FastAPI, Python, Machine Learning, APIs | GitHub

- Led a four-member team in developing an EV route and charging optimization system during Vihaan 9.0.
- Developed machine learning models for charging wait-time prediction and intelligent charging-stop recommendations.
- Integrated backend APIs, optimization algorithms, and ML pipelines into a unified FastAPI backend.
- Advanced to Round 3 of Vihaan 9.0 through iterative development and system refinement.

ACHIEVEMENTS

- 3rd Place – Build with Gemma Hackathon (Google & AIMS)**

Jul 2026
- Led a 4-member team to secure **3rd Place** in the AI for Legal Assistance Track by developing **Legal.AI**, a multilingual citation-grounded legal assistant powered by Gemma 4 and a hierarchical RAG pipeline.

TECHNICAL SKILLS

- Languages:** Python, Java, C, C++, JavaScript, HTML, CSS
- Machine Learning & AI:** Scikit-learn, XGBoost, LangGraph, Retrieval-Augmented Generation (RAG), Sentence Transformers, Google Gemma
- Frameworks & Technologies:** FastAPI, React, Node.js, Firebase, Tailwind CSS
- Databases:** ChromaDB, MongoDB, Cloud Firestore, SQLite
- Tools:** Git, GitHub, VS Code, Jupyter Notebook, Vercel

CERTIFICATIONS

- Machine Learning Specialization**

Coursera
- Intermediate Machine Learning**

Kaggle
- Intro to Machine Learning**

Kaggle
- AI Python for Beginners**

DeepLearning.AI
- Ultimate Web Development Course 2026**

Udemy

RELEVANT COURSEWORK

- Data Structures and Algorithms, Object-Oriented Programming, Software Engineering, Machine Learning

CLUBS & ACTIVITIES

- Rotaract Club of DTU Regency**

Jul 2026 – Present

Director of Rotary Relations
- Rotaract Club of DTU Regency**

Sep 2025 – Jul 2026

Member
- DTU Literature & Film Council**

Aug 2025 – Present

Corporate Executive
- Mathematics & Computing Society (MACS)**

Jan 2026 – Present

DSA-CP Team Member